

Christian Carrico

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PROJECTS

AFSIM Reinforcement Learning Environment

- Designed and delivered solutions for a Python Reinforcement Learning environment that integrates AFSIM and graph based observations to optimize blue force outcomes
- Developed code to procedurally generate sufficiently random AFSIM scenarios, providing a reinforcement learning agent with enough experiences to enhance generalization across the entire problem space
- Built Regex- and Jinja2-powered scripts to parse hundreds of AFSIM model files into a Python interface, enabling better scenario generation and a richer feature space
- Optimized Docker and Docker Compose configurations, reducing environment development image size by 60%, dramatically reducing build times and accelerating deployments to a classified network

Modeling & Simulation Framework Trade Study

- Directed research and assessment of multiple M&S frameworks, coordinating tasks with government and contractor personnel
- Developed objective evaluation criteria and formalized use cases to determine the most suitable framework
- Compiled findings into a comprehensive report and presented key insights to leadership for decision-making

Workflow Planning Web Application

- Led the development of a COCOM exercise planning tool using Agile methodologies by integrating feedback from Warfighters into tasks for the next sprint
- Enhanced codebase quality by refactoring C# controllers and standardizing JavaScript web views, improving maintainability and simplifying feature delivery

EXPERIENCE

Combat Capability Development Command

Colorado Springs, CO

Software Engineer

Apr 2018 – Present, Full-time

- Drove progress of multiple Python projects by facilitating planning discussions and tasking government developers and contractors across various organizations and companies
- Employed Amazon Web Services to deploy critical infrastructure such as CI/CD pipelines and collaboration tools to support various government efforts
- Leveraged technologies such as the Atlassian tool suite and GitLab to capture customer requirements, document design decisions, and accelerate the development process
- Partnered with small businesses on SBIR/STTR contracts, overseeing progress and deliverables while providing technical guidance to maximize government value
- Utilized SysML and Cameo to develop system models, supporting early-stage architecture design, analysis, and CONOPs
- Mentored student interns and new employees by delegating meaningful tasks that foster skill development and professional growth
- Evaluated research proposals for technical and commercial merit while ensuring alignment with Agency priorities

Electrical & Computer Engineering College

Fort Collins, CO

Student Ambassador

Aug 2012 – Aug 2014, Part-time

- Engaged prospective students during recruitment events to promote interest in the college's programs and drive enrollment growth

Colorado State University Advanced Beam Laboratory

Fort Collins, CO

Intern

Aug 2012 – July 2014, Full-time/Part-time

- Designed and managed PHP website while curating content to showcase laboratory research and personnel

EDUCATION

Colorado State University

Fort Collins, CO

B.S. in Computer Engineering

Aug 2012 – Dec 2017

B.S. in Mathematics, Concentration in Information Security

Aug 2012 – Dec 2017